LASIGE data and systems intelligence

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Deep Learning-based Digital Twin of Manufacturing Systems

Speaker
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Affiliation
G-SCOP @ University of Grenoble and LASIGE

When
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Where
C6.3.27

Abstract

Digital Twins (DT) are key technologies in Industry 4.0, providing a powerful tool for data-driven decision-making, predictive maintenance, and operational optimization. Yet, challenges arise in creating accurate DT that mirror the complexities of real-world manufacturing systems. This talk explores the concept of DT in Industry 4.0, using the Tidal Turbine Production as case study and focusing on the assembly and disassembly processes. The discussion will cover specific challenges of this use case and a research methodology based on computer vision and deep learning algorithms. Preliminary results and insights into the next steps, particularly regarding model integration and real-time data exchange, will also be shared.

Short Bio

Nessrine FARHAT is a PhD student pursuing a co-tutorship thesis between the University of Lisbon at LASIGE and the University Grenoble Alpes at the G-SCOP laboratory. Her research focuses on Deep Learning-based Digital Twin of Manufacturing Systems. She holds a MSc degree in Data Exploration and Decision Support from the Galilée Institute, Sorbonne Paris Nord, in collaboration with the Web Intelligence and Data Science Master's program at the Faculty of Science, USMBA, Morocco.

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